SDS 320E – Descriptive Analysis Report

Write in complete sentences using single-spaced, 12-pt font. Include figures in text. ***3-page limit.***

Name: ***Nicholas Hinkel***
EID: ***nsh748***

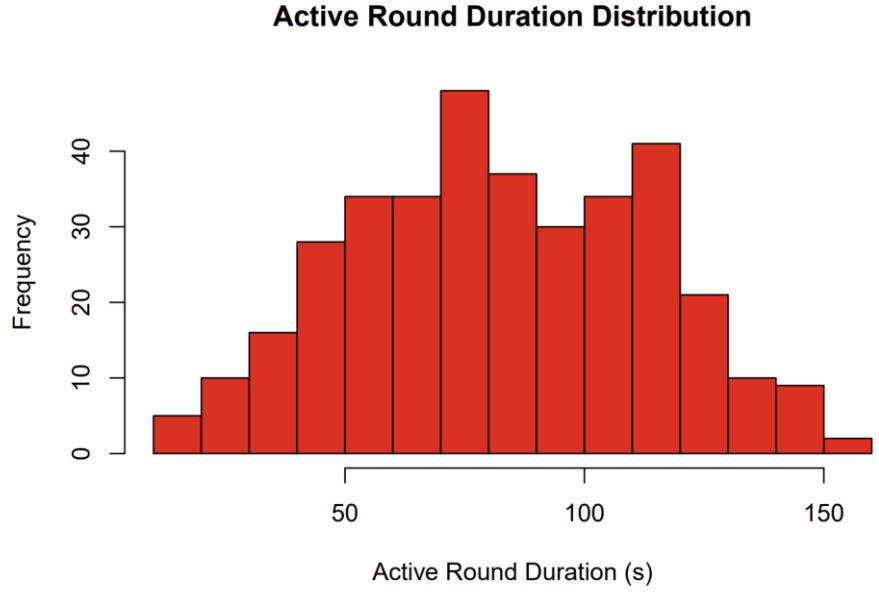
Introduction

The goal of this study is to analyze data from professional CS2 matches to compare round duration to when the opening kill of the round occurs and which side wins the round. I retrieved my data from the OpenCS2 dataset that contains statistics from professional CS2 matches. The sampling units are single rounds chosen randomly from a match. My final sample size is 359 rows. Of the 16,527 rows, 16,168 rows were removed to use only 1 round from each match and to remove rows with missing data. One research question is to find how round duration associates with the amount of time until the first kill. The other research question is to find how the round duration associates with the winning side of the round. The round duration used for this analysis was recalculated to exclude the buy period at the beginning of the round (active round duration).

Research Questions

RQ1 – How does the round duration associate with the amount of time until the opening kill?
RQ2 – How does the round duration associate with the winning side of the round?

Descriptive Analysis of Response Variable *(include graph of distribution here)*

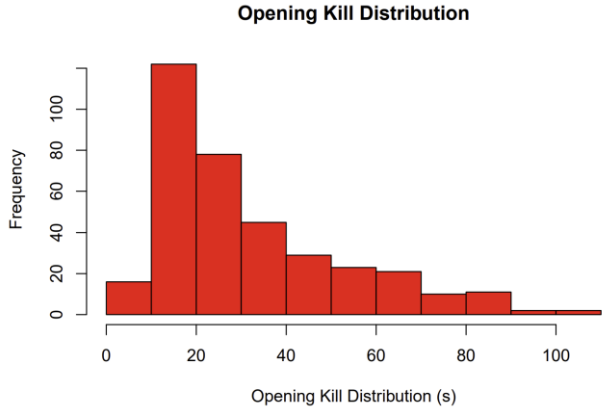


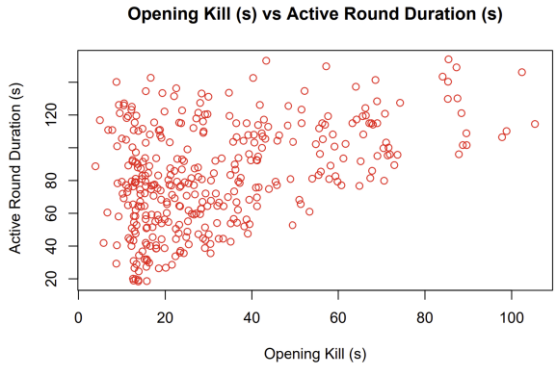
Active Round Duration appears to have a mostly normal distribution with a mean of 82.68 seconds and a standard deviation of 30.78 seconds.

SDS 320E – Descriptive Analysis Report

Write in complete sentences using single-spaced, 12-pt font. Include figures in text. ***3-page limit.***

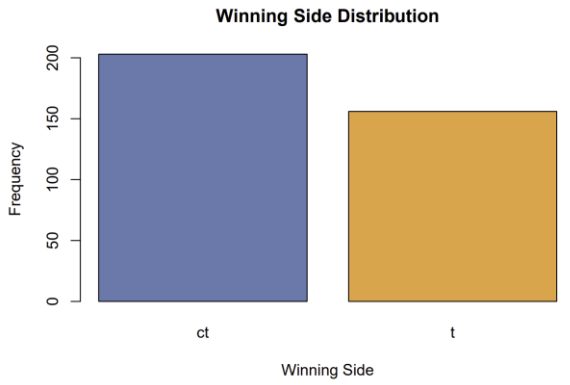
Investigation of Numeric Explanatory Variable *(include univariate, bivariate graph here)*

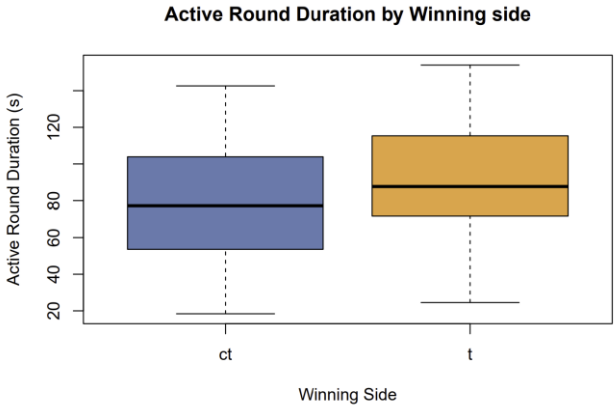




Opening kill has a right-skewed distribution with a median of 25.13 seconds, an IQR of 25.82 seconds and possibly some outliers. The scatterplot displays a positive association between the opening kill and the active round duration.

Investigation of Categorical Explanatory Variable *(include univariate, bivariate graph here)*





The barplot shows that CT had a greater win amount of 203 wins (56.55%) than T had with 156 wins (43.45%). The grouped boxplot displays that CT and T have a similar spread, with T having a slightly higher median.

SDS 320E – Descriptive Analysis Report

Write in complete sentences using single-spaced, 12-pt font. Include figures in text. ***3-page limit.***

R Code *(organized by variable without output or extraneous syntax)*

```
#Descriptive Analysis of Response Variable:
hist(df$active_round_duration_s,
    main = "Active Round Duration Distribution",
    xlab = "Active Round Duration (s)",
    ylab = "Frequency",
    col = "#FF0000")
summary(df$active_round_duration_s)
sd(df$active_round_duration_s)

#Investigation of Numeric Explanatory Variable:
hist(df$opening_kill_seconds,
    main = "Opening Kill Distribution",
    xlab = "Opening Kill Distribution (s)",
    ylab = "Frequency",
    col = "#FF0000")
summary(df$opening_kill_seconds)
sd(df$opening_kill_seconds)
IQR(df$opening_kill_seconds)
plot(df$active_round_duration_s ~ df$opening_kill_seconds, data = df,
    main = "Opening Kill (s) vs Active Round Duration (s)",
    xlab = "Opening Kill (s)",
    ylab = " Active Round Duration (s)",
    col = "#FF0000")

#Investigation of Categorical Explanatory Variable:
winner_table <- table(df$winner_side)
barplot(winner_table,
       main = "Winning Side Distribution",
       xlab = "Winning Side",
       ylab = "Frequency",
       col = c("#5d79ae", "#eda338"))
winner_table
prop.table(winner_table)
boxplot(df$active_round_duration_s ~ df$winner_side, data = df,
       main = "Active Round Duration by Winning side",
       xlab = "Winning Side",
       ylab = "Active Round Duration (s)",
       col = c("#5d79ae", "#eda338"))
```

AI used for assistance